



Presentation 2: Reactors

Adil, Kyle, Miguel and Omar

Introduction

Importance

- Found in all nearly all chemical processes
- Necessary for controlling reactions
- Different processes require different reactors
- Ensures consistent quality
- Can be modified to achieve desired results

Categories

- Flow based
- Batch Based
- Designed by volume
- Designed by catalyst weight
- Specialty reactors
- Adiabatic
- Isothermal

Batch Reactor



Design and Operating Conditions

- Closed vessel; reactants loaded, reacted, then discharged
- Equipped with agitation and heating/cooling jacket
- Operates under unsteady-state conditions.

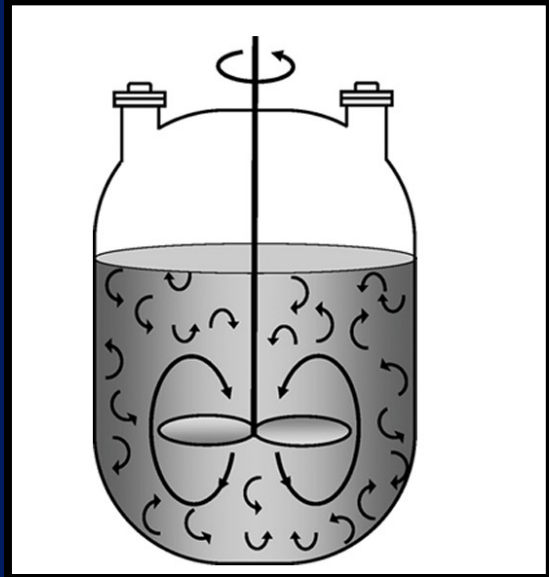


Figure 1 Batch Reactors (McGrawHill, 1956)

Rules of Thumb

- Good for making different products
- Best for small-scale, or high-value chemicals
- Easy to operate, but not economical for large scale processes

Applications

- Common reactions
 - Polymer synthesis
 - Fermentation
- Common in research scale production
- Common industries
 - Pharmaceutical
 - Biochemical
 - Fine-chemical
- Ideal when high purity is required.

Continuous Stirred Tank Reactor (CSTR)



Design and Operating Conditions

- Continuous feed and product streams at steady state
- Well-mixed (uniform concentration and temperature)
- Outlet composition is the same as inside CSTR
- Best for liquids

Rules of Thumb

- Good mixing is required
- Multiple CSTR in series increases conversion
- Easier to scale up than batch systems

Applications

- Fermentation
- Neutralization
- Polymerization
- Wastewater Treatment
- Ammonium nitrate and acetic acid production
- Ideal for liquid-phase reaction necessitating constant control

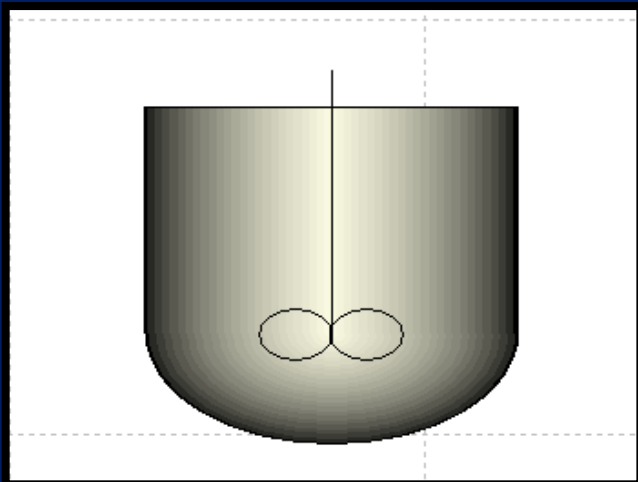


Figure 2 CSTR PFD Symbol (ASPEN)

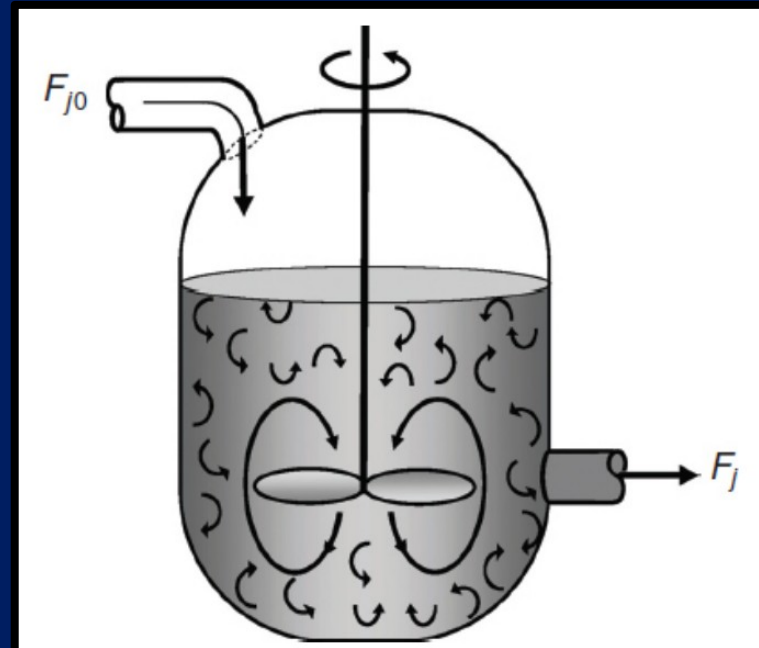


Figure 3 CSTR (Folger, 2016)

Fluidized Bed Reactor (FBR)



Design and Operating Conditions

- Upward gas flow or liquid suspended catalyst
- Well mixed and uniform temperature
- Cyclone separator typically necessary

Rules of Thumb

- Utilize Ergun equation to calculate minimum fluidization velocity and operate above
- Can be modeled as a CSTR with catalyst

Applications

- Fluid catalyzed cracking (FCC)
- Coal combustion
- Chlorination
- Polymerization
- Highly exothermic process

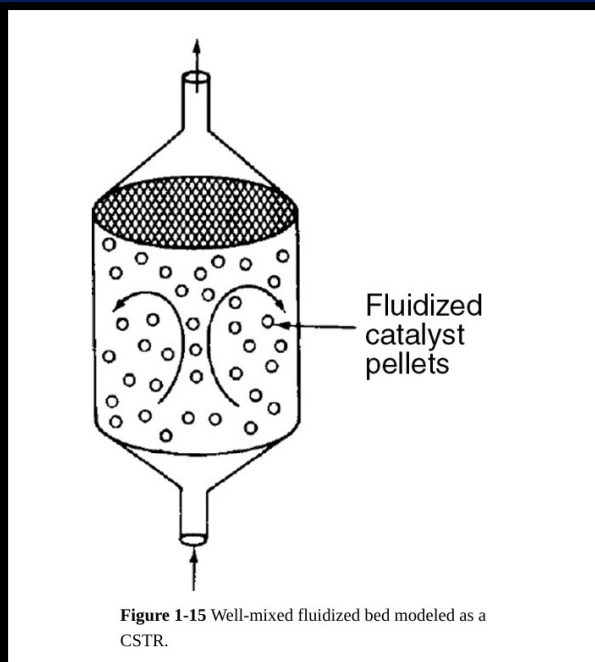


Figure 4 Fluidized bed reactor
(Folger, 2016)

Plug Flow Reactor (PFR)

Design and Operating Conditions

- Plug flow
- Gas Phase or fast liquid phase reactions
- Varying of concentration and temperature with length

Rules of Thumb

- Typically, tubular $L/D > 50$
- High Conversion for a single pass

Applications

- Gas phase oxidations
- Methanol synthesis
- Nitration
- Steam reformation
- Common Industries
 - Petrochemical
 - Polymer
- Preferred for fast high temperature reactions

Plug flow—no radial variations in velocity, concentration, temperature, or reaction rate

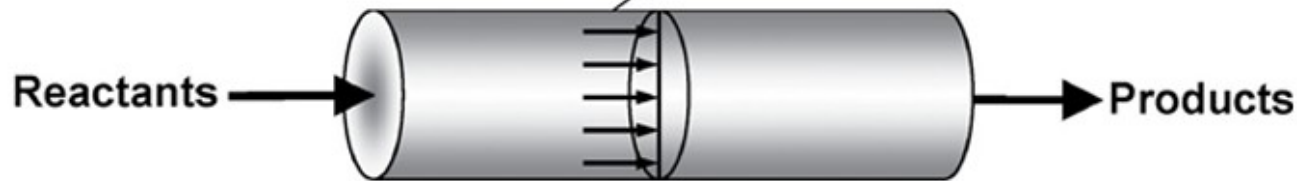


Figure 5 Plug flow Reactor (Foglers, 2016)

Packed Bed Reactor (PBR)

Design and Operating Conditions

- Reactants pass through catalysts
- Temperature and conversion vary along the length

Rules of thumb

- Catalyst sizes:
 - 3-6 mm for gas phase
 - 1-3 mm for liquid phase

Applications

- Ammonia Synthesis
- Hydrogenation
- Oxidation
- Common Industries
 - Refining
 - Chemical Manufacturing
 - Fertilizer

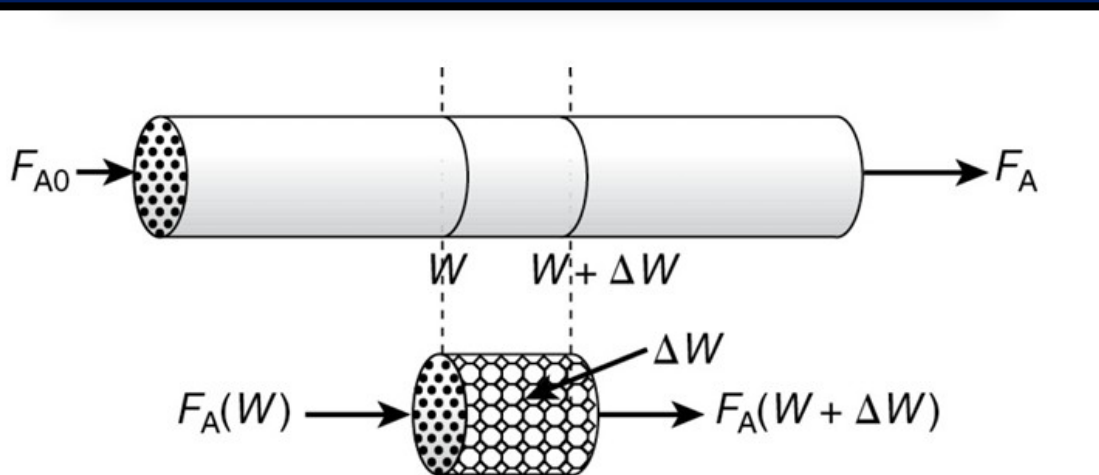


Figure 1-14 Packed-bed reactor schematic.

Figure 6 Packed bed Reactor (Folger, 2016)

Semi-Batch Reactor

Design and Operating Conditions

- One or more reactants added gradually due to their gaseous or exothermic nature

Rules of Thumb

- Feed rate is adjusted for temperature control and selectivity
- Residence time depends on inlet rate and volume

Applications

- Polymerization
- Neutralization
- Hydrogenation
- Biochemical
- Ideal for exothermic and gas phase reactions
- Common Industries
 - Fine-chemical
 - Bio-tech

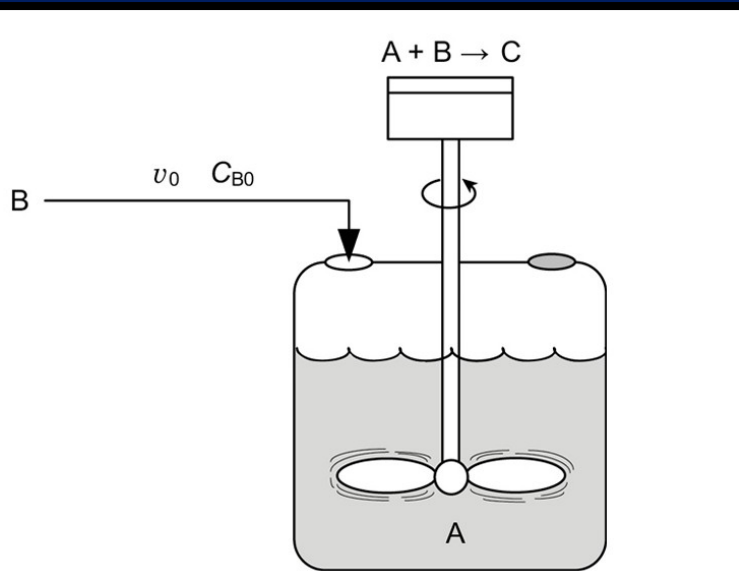


Figure 7 Semi-batch Reactor (Folger, 2016)

Membrane Reactor

Design and Operating Conditions

- Semi-permeable membrane allows for selective separation
- Used to shift equilibrium by removing product

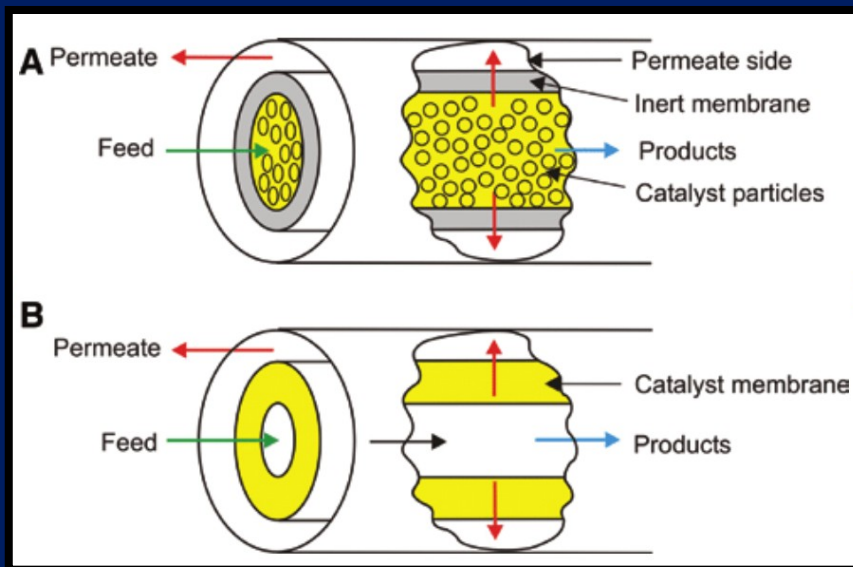


Figure 8 Membrane Reactor (Kiel, 2017)

Rules of Thumb

- Useful for parallel reactions or equilibrium limited reactions
- Use moderate pressure differentials
- To balance reaction rate and separation efficiency, optimize membrane area to volume ratio

Applications

- Hydrogen Production, methane reforming, dehydrogenation, CO_2 reduction
- Used in energy and environmental technologies
- Promotes process intensification and efficiency

Conclusion

Key Takeaways

- Choice of reactor is important
- Operating conditions guide reactor choice
- Incorrect reactor choices can lead to hazardous operations



Figure 9 Blown up Reactor in India (Sharma, 2025)

Bibliography

Folger, H. S. (2016). *Elements of Chemical Reaction Engineering* . Pearson Education.

Kiel, F. J. (2017). Process Intensification. *Reviews in Chemical Engineering*, 17.

McGrawHill. (1956). *Chemical Engineering* 63, 200.

Mettler Toledo. (n.d.). Continuous Stirred Tank Reactors (CSTRs). *Flow Technology for Chemical and Biological Syntheses*.

Sam Catalano, A. W. (n.d.). Visual Encyclopedia of Chemical Engineering Equipment. *Plug Flow (PFR)*.

Sharma, S. (2025, July 1) At least 26 dead after chemical reactor explodes in drug factory in India. *The Independent*
<https://www.independent.co.uk/asia/india/telangana-chemical-factory-blast-sigachi-death-toll-b2780102.html>